

2	(a) amount of substance containing same number of particles as in 0.012 kg of carbon-12	M1 A1	[2]
	(b) $pV = nRT$ amount = $(2.3 \times 10^5 \times 3.1 \times 10^{-3}) / (8.31 \times 290)$ + $(2.3 \times 10^5 \times 4.6 \times 10^{-3}) / (8.31 \times 303)$ = $0.296 + 0.420$ = 0.716 mol (give full credit for starting equation $pV = NkT$ and $N = nN_A$)	C1 C1 C1 A1	[4]

3	(a) charges on plates are equal and opposite	M1
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